

Fully Passive Quantum Key Distribution

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(Received 29 November 2022; accepted 19 April 2023; published 31 May 2023)

We propose a fully passive linear optical quantum key distribution (QKD) source that implements both random decoy-state and encoding choices with postselection only, thus eliminating all side channels in active modulators. Our source is general purpose and can be used in, e.g., BB84, the six-state protocol, and reference-frame-independent QKD. It can even potentially be combined with measurement-device-independent QKD to achieve robustness against side channels in both detectors and modulators. We also perform a proof-of-principle experimental source characterization to show its feasibility.

DOI: [10.1103/PhysRevLett.130.220801](https://doi.org/10.1103/PhysRevLett.130.220801)

Background.—Quantum key distribution (QKD) [1,2] offers, in principle, information-theoretic security for communication between two users, Alice and Bob. However, practical imperfections including side channels may lead to security loopholes [3–8]. While measurement-device-independent QKD [9] removes all side channels from the detectors, *source* imperfections remain a challenge to the practical security of QKD systems. For instance, modulators may introduce side channels [5,6] and are susceptible to Trojan horse attacks [3,4] from Eve.

Passive decoy-state setups [10,11] and a passive encoding scheme [12] have both been proposed for QKD, which respectively perform the decoy-state [13–15] intensity choice and the BB84 state choice by only performing postselection on measurement results, thus avoiding the use of source modulators [16]. An illustration of them can be found in Figs. 1(a) and 1(b).

Nonetheless, up to now, for more than 10 years, passive encoding has never been successfully combined with passive decoys with linear components and weak coherent pulse (WCP) sources. The main challenge is this: *the intensity and polarization of the prepared states are coupled in a passive QKD setup*. This makes passive-encoding QKD incompatible with the standard decoy-state analysis, thus severely limiting its practical use.

Our contribution.—In this Letter, we make two main contributions: (1) we design a *passive source* capable of creating a coherent state of arbitrary polarization and arbitrary intensity with only linear optics. (2) We propose

a class of *passive QKD protocols* using our new source and design a set of postselection strategies to decouple the intensity and polarization distributions and enable decoy-state analysis.

This work allows *fully passive* QKD with both encoding and decoy setting choices implemented passively via local detection and postselection only, hence eliminating all side channels in the source modulators. Importantly, our source is *general purpose* and can be used in various protocols. Indeed, the class of passive protocols we propose in this work is only one example. Our approach can also potentially be combined with measurement-device-independent (MDI) protocols to further eliminate detector side channels and enable even higher implementation security. We believe that this work will open up a whole new direction of highly practical fully passive QKD systems.

Some alternative approaches worth noting include Refs. [24–26]. A brief discussion of the difference between our proposal and each of them is included in the Supplemental Material [27], Sec. A.

Fully passive source.—As shown in Fig. 1(c), we propose a setup where Alice uses four laser sources operating at the strong light level. Importantly, we assume *all lasers are independently phase-randomized*, the justification of which will be discussed in a later section. *Two pairs of lasers respectively interfere* in a setup similar to that of passive decoy-state generation, *outputting signal pulses with arbitrary intensities μ_H and μ_V* (between 0 and $\mu_{\max}$, the sum of intensities from one pair of lasers) *and random phases ϕ_H*

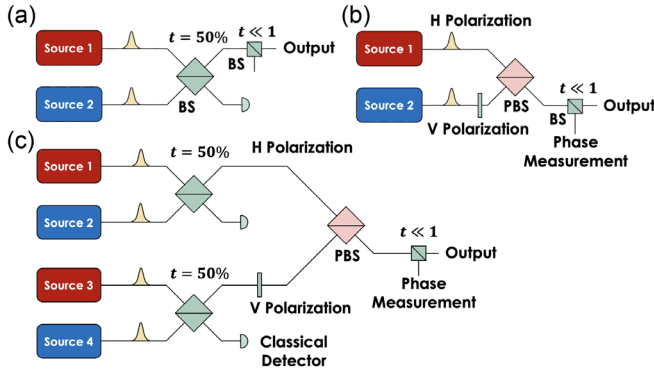


FIG. 1. (a) Passive decoy-state setup [10,11]. The output intensity is determined by the random phase difference between the two sources. Alice postselects the intensity with a classical detector to prepare decoy-state settings. (b) Passive encoding setup [12]. The output polarization is determined by the random phase difference between two sources. Alice postselects the polarization with a phase measurement to prepare BB84 states. (c) Our fully passive source. The output state is a phase-randomized coherent state with arbitrary polarization and arbitrary intensity. Alice can postselect based on observations from her two classical detectors and her phase measurement between the H and V modes. Here, the sources emit *strong* light, which is attenuated to the single-photon level before being sent out. The same setup is applicable to time-bin phase encoding if we replace the polarizer and the polarizing beam splitter with a delay line and a beam splitter.

and ϕ_V . The pulses are rotated into H and V polarizations and combined at a polarizing beam splitter. The output state is still a coherent state, with intensity $\mu = \mu_H + \mu_V$ and a global random phase (say ϕ_H [40]). The polarization mode can be described by the creation operator

$$a_{\theta_{HV}, \phi_{HV}}^\dagger = \cos(\theta_{HV}/2) a_H^\dagger + e^{i\phi_{HV}} \sin(\theta_{HV}/2) a_V^\dagger, \quad (1)$$

where we have defined $\theta_{HV} = 2 \cos^{-1}[\sqrt{\mu_H/(\mu_H + \mu_V)}]$ and $\phi_{HV} = \phi_V - \phi_H$ (the polar and azimuthal angles that uniquely determine a state on the Bloch sphere). This means that, given a set of input parameters $(\mu_H, \mu_V, \phi_H, \phi_V)$, we can always map it to some phase-randomized coherent state with some polarization on the Bloch sphere. In fact, even the inverse holds true (a proof can be found in the Supplemental Material, Sec. B), i.e., any desired intensity within $(0, \mu_{\max})$ [41] and any polarization on the Bloch sphere can be created by some parameters $(\mu_H, \mu_V, \phi_H, \phi_V)$. This means that our source is general purpose and is capable of creating a coherent state with arbitrary intensity and polarization on the Bloch sphere while also maintaining global phase randomization.

Passive protocol.—To use the source, Alice needs to use the observations from her classical measurements to decide on a set of postselection strategies to determine the states she intends to prepare. Here, as an example, we show how she can prepare six polarization states $\{H, V, +, -, L, R\}$ in three bases and perform the decoy-state analysis. Note that

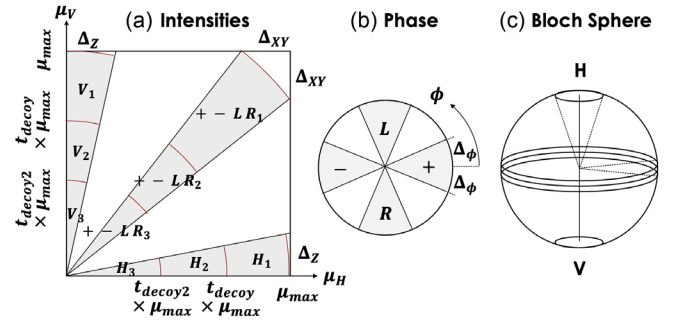


FIG. 2. (a) Postselection on intensities (μ_H, μ_V) . The shaded regions correspond to the Z basis and X - Y plane on the Bloch sphere. The slope $x = \mu_V/\mu_H$ on this plot determines the latitude (polar angle) θ_{HV} of the state on the Bloch sphere. The postselected regions can further be divided into subregions to implement decoy states. (b) The phase postselection step that determines the longitude of the state, which helps Alice determine, e.g., the X basis and Y basis polarization states. (c) Corresponding states on the Bloch sphere based on the intensity postselection step.

this is not her only possible choice, but we select this set of states as they can be used in a wide class of symmetric [42] protocols such as BB84, reference-frame-independent (RFI) QKD, and the six-state protocol. A pair of such sources can even in principle be used for MDI-QKD or RFI-MDI-QKD.

Alice can perform two types of postselection: (1) Alice observes $(\mu_H, \mu_V, \phi_{HV})$ from her measurements and defines postselection regions $\{S_i\}$ on these variables. Signals that fall within a region are considered to be prepared in the given state. An example set of $\{S_i\}$ can be found in Fig. 2, which allows us to prepare states in six polarizations $\{H, V, +, -, L, R\}$, and also prepare various decoy settings using the subdivided regions. (2) Importantly, Alice can perform an *additional postselection* on the signals to either keep or discard signals based on an arbitrary distribution $q_\mu(\mu_H, \mu_V)$. This is equivalent to a coin flip whose bias depends on the values (μ_H, μ_V) that Alice measures. This allows Alice to shape the natural intensity probability distribution resulting from the passive decoy interferometers $p_\mu(\mu_H, \mu_V) = 1/[\pi^2 \sqrt{\mu_H(\mu_{\max} - \mu_H)\mu_V(\mu_{\max} - \mu_V)}]$ into an arbitrary distribution $p'_\mu = p_\mu \times q_\mu$ she desires. For this, Alice requires some local random bits, which could be obtained from a quantum random number generator. This strategy can be used in our decoy-state analysis to decouple the distribution of polarization from intensities. An illustration can be found in Fig. 3.

A main difference between a passive QKD protocol and an active one is that, in the former, all states Alice prepares are postselected over finite regions $\{S_i\}$. This results in two problems: (1) each of the decoy settings represents a collection of WCP states with a continuous range of (i.e., mixed) polarizations and intensities. We need to use these collections of states to estimate the single-photon statistics. (2) The single-photon component in the signal

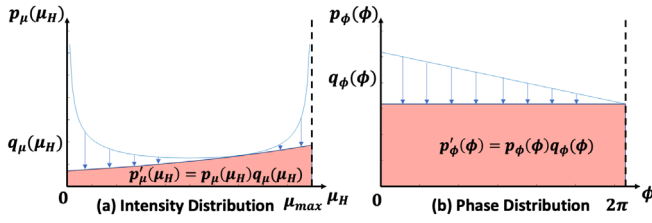


FIG. 3. Examples of additional postselection on intensity or phase distribution by Alice. We can use this technique to, e.g., modify the intensity distributions of μ_H or μ_V , or to shape a nonuniform but known phase distribution into a uniform one.

state, say in the Z basis, has mixed polarizations too. This is an encoding imperfection and requires additional security analysis. (Ref. [44] studied a similar imperfection, although its analysis is only applicable to MDI-QKD.)

We address the problem in two steps: (1) we design specific postselection regions that allow us to perform standard decoy-state analysis and obtain the upper and lower bounds on the statistics for perfectly prepared single photons, $\{Y_1^{\text{perfect}}, e_1 Y_1^{\text{perfect}}\}$. (2) We show that the mixed polarization in the single-photon state preparation does not increase the privacy amplification amount needed, i.e., we can simply use the statistics from perfectly encoded photons to calculate the key rate.

First, let us look at the decoy states. Because of the finite postselection regions, in order to estimate the yields (a similar argument applies to the error yields), the linear constraints take the form of

$$\langle Q \rangle_{S_i} = \sum_{n=0}^{\infty} \langle P_n Y_n \rangle_{S_i}, \quad (2)$$

where the brackets represent integrating over S_i and calculating the expected value, while Q , P_n , and Y_n are respectively the gain, the Poissonian distribution, and the n -photon yield.

The biggest challenge is that the decoy states have mixed and correlated polarizations and intensities, resulting in P_n and Y_n being coupled, as both P_n and Y_n depend on the polarization of the signal (which takes a range of values in S_i). This prevents us from constructing a linear program for the decoy analysis. To solve this problem, we prove that, if we use the particular “sector-shaped” regions in Fig. 2(a), and use postselection to shape the intensity distribution into $p_\mu \propto \exp(\mu_H + \mu_V)$, we can decouple the polarization distribution from the average photon number distribution and rewrite Eq. (2) as

$$\langle Q \rangle_{S_i} = \sum_n \langle P_n \rangle_{S_i} \times Y_n^{\text{mixed}}, \quad (3)$$

where Y_n^{mixed} is the yield Y_n integrated over all possible polarization angles (θ_{HV}, ϕ_{HV}) in S_i following an angular probability distribution. Importantly, Y_n^{mixed} is in general different for each S_i (meaning that a linear program cannot

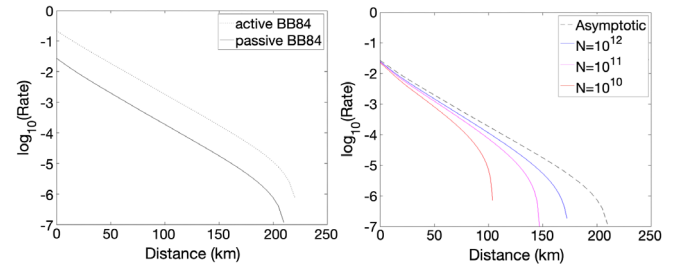


FIG. 4. Left: key rate comparison between active and fully passive decoy-state BB84 assuming infinite data size, using the strategy in Fig. 2. The size of the Z basis postselection regions and the maximum intensities are optimized. Right: key rate simulation for fully passive decoy-state BB84 under finite-size effects (considering collective attacks) with different data sizes. A full optimization on all postselection parameters is performed.

be constructed), but our specific postselection strategy allows Y_n^{mixed} to be the same for all decoy settings S_i . A rigorous proof can be found in the Supplemental Material, Sec. C. Equation (3) can be solved as a linear program to obtain the lower bound $Y_1^{\text{mixed},L}$ (and similarly the upper bound $e_1 Y_1^{\text{mixed},U}$ for the error yield). We also prove that, specifically for single photons, $Y_1^{\text{mixed},L}$ and $e_1 Y_1^{\text{mixed},U}$ happen to respectively also be lower and upper bounds on the yield and error yield Y_1^{perfect} and $e_1 Y_1^{\text{perfect}}$ of single-photon states prepared with perfect polarizations.

We further show that the mixed polarizations in the single-photon components of the signal state do not increase the amount of privacy amplification needed. In simple words, if we consider the equivalent entanglement distribution picture, the imperfect single-photon state preparation is equivalent to noisy positive operator-valued measures (i.e., classical postprocessing noise) in Alice’s system. This means that Eve cannot gain additional information. A detailed proof can be found in the Supplemental Material, Sec. D. We can then simply use the aforementioned perfect statistics to estimate the privacy amplification amount when calculating the key rate, i.e., $Y_1^{\text{perfect},Z} [1 - h_2(e_1^{\text{perfect},X})]$.

Simulation results.—We include numerical simulation results for the fully passive decoy-state BB84 protocol in Fig. 4. A simulation for passive RFI-QKD can also be found in the Supplemental Material, Sec. H. We set a misalignment of $e_d = 2\%$ on the X-Z plane, per-detector dark count $p_d = 10^{-6}$, detection efficiency of 1 (merged into channel loss), error-correction efficiency of $f_e = 1.16$, and a failure probability of $\epsilon = 10^{-7}$ for the finite-size scenario. We see that, asymptotically, the fully passive protocol has reasonable performance, albeit having about a 1 order of magnitude lower key rate compared to its active counterpart, mainly due to the sifting factor resulting from postselection and the inherent quantum bit error rate due to using finite postselection regions. We can also see that, despite extensive postselection, the protocol has quite good

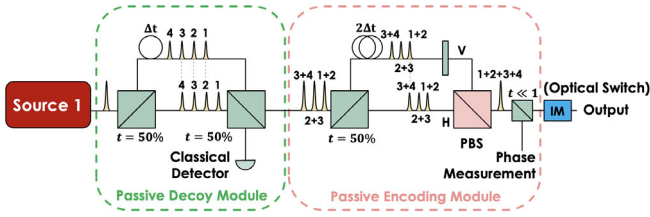


FIG. 5. An equivalent setup for a fully passive source using only one single laser. This setup makes use of two delay lines of Δt and $2\Delta t$ to interfere four independent pulses from different time bins. Here, we need to use a fixed-intensity modulator (or an optical switch) to suppress every three out of four output pulses. Note that this does not leak additional information as a fixed pattern of signals is selected and no random number is used [56].

performance in the finite-size scenario, even with a relatively small data size like $N = 10^{10}$ sent by Alice.

Experimental considerations.—Our proposed scheme depends on three key assumptions: (1) the phase distributions from the sources are random, uniform, and independent. (2) Alice can accurately measure the intensities and the phase of each pulse. (3) The four laser pulses can be maintained at the same intensities, polarizations, and frequencies for high interference visibility. Assumption 1 is in fact a prerequisite for active QKD, too, and there have been successful implementations of QKD systems [45] relying only on the phase randomness of the source without active phase randomization, as well as usage of such randomness in quantum random number generators [46–50]. Assumption 2 imposes requirements on Alice’s measurement devices, where inaccuracies resulting in imperfect characterization of $\langle P_n \rangle_{S_i}$ in the decoy-state analysis would have similar effects as intensity fluctuations in active QKD. Assumption 3

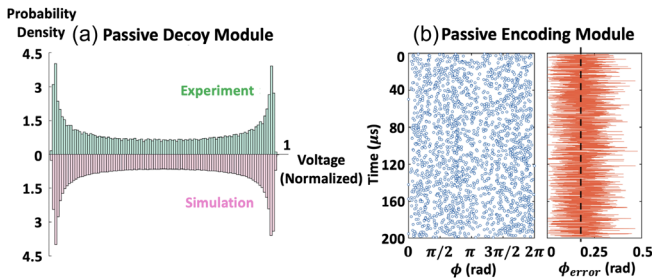


FIG. 6. (a) Histogram of experimentally observed intensities for the passive decoy module. It follows the theoretical prediction well, implying uniform phase distribution, good measurement accuracy, and good interference visibility. Additionally, the intensity data passed an autocorrelation test, implying randomness of phase distribution. (b) Measured phase ϕ and characterized error ϕ_{error} , showing we can perform accurate phase measurement. Phase measurements are performed by measuring in the X or Y bases (detailed setup included in the Supplemental Material, Sec. J). For simplicity, the passive encoding module is separately tested from the passive decoy module, meaning that here we only create and measure states randomly on the equator of the Bloch sphere.

may pose a bigger challenge for experimentalists. While interference from off-the-shelf WCP sources has been reported [52–55], frequency mismatch in our setup will result in phase drift, so one would need to choose narrow laser bandwidths and short pulse widths. We also propose an alternative setup in Fig. 5, where we use a single laser, delay lines, and an optical switch to implement the source and avoid frequency mismatch.

We have performed a proof-of-principle experimental characterization of our source to show its feasibility. We build our system based on Fig. 5, which sequentially consists of a single gain-switched laser, a passive decoy module, and a passive encoding module. In Fig. 6 we show that the source has random and uniform phase distribution and that we can perform accurate intensity and phase measurements.

Discussion.—In this Letter, we have presented a simple yet effective scheme to implement a fully passive QKD protocol and applied it to passive decoy-state BB84 (and to RFI-QKD in the Supplemental Material). Our source is in principle also compatible with MDI-QKD, and a similar postselection idea can even be applied to twin-field QKD sources.

We thank Yue Jiang, Kai Sum Chan, Chenyang Li, Yujia Zhou, and Xiaohao Chen for helpful discussions. This project is financially supported by the University of Hong Kong start-up grant and NSERC. H.-K. L. also acknowledges support from MITACS, CFI, ORF, Huawei Technologies, Inc., the Royal Bank of Canada, and Innovative Solutions Canada. W. W. acknowledges support from the Hong Kong RGC General Research Fund (GRF) and the University of Hong Kong Seed Fund for Basic Research for New Staff. M. C. and V. Z. acknowledge support from the Galician Regional Government (consolidation of Research Units: AtlantTIC), the Spanish Ministry of Economy and Competitiveness (MINECO), the Fondo Europeo de Desarrollo Regional (FEDER) through Grant No. PID2020-118178RB-C21, MICIN with funding from the European Union NextGenerationEU (PRTR-C17.I1) and the Galician Regional Government with own funding through the “Planes Complementarios de I+D+I con las Comunidades Autonomas” in Quantum Communication, and the European Union’s Horizon Europe research and innovation programme under the project QSNP (Quantum encryption and future quantum network technologies).

Note added.—Recently, an alternative approach to passive QKD has been proposed [60].

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- [41] In fact, the intensity can exceed $\mu_{\max}$ (up to $2\mu_{\max}$ when θ_{HV} is exactly $\pi/2$), at the expense of a more limited range of polarizations.
- [42] Note that, here in our protocol definition and decoy-state analysis we focus on a set of symmetric protocols, which prepare symmetric pairs of states for each basis and for, e.g., the X and Z bases the single-photon states satisfy $\rho_1^Z = \rho_1^X$ (i.e., they have basis independence). For asymmetric protocols, such as the three-state loss-tolerant protocol [43], our source would still be able to generate the desired states, but the decoy-state analysis would require a case-by-case revision. Nonetheless, for, e.g., loss tolerance, one can consider as well other symmetric protocols, such as those using the approach in Ref. [44], to avoid the asymmetry in the protocol.
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